IBM HR Employee Attrition — Data Preparation & Cleaning Log

Source sheet: WA_Fn-UseC_-HR-Employee-Attriti | Table: HR_Attrition_Data | 1,470 employees, 35 source columns + 5 helper columns

1. Dataset overview & key uniqueness

Check	Result	Status
Total rows	1,470	**OK**
Total columns (incl. helper columns)	40	**OK — 35 source + 5 helper**
Distinct EmployeeNumber values	1,470	**OK — one row per employee**
Duplicate EmployeeNumber rows	0	**OK — no duplicates**
Blank cells across entire table	0	**OK — no blanks**

2. Column-level checks — data type, blanks & variance

Column	Expected type	Blanks	Numeric cells	Text cells	Distinct values	Verdict
Age	Number	0	1470	0	43	**OK**
Attrition	Text	0	0	1470	2	**OK**
BusinessTravel	Text	0	0	1470	3	**OK**
DailyRate	Number	0	1470	0	886	**OK**
Department	Text	0	0	1470	3	**OK**
DistanceFromHome	Number	0	1470	0	29	**OK**
Education	Number	0	1470	0	5	**OK**
EducationField	Text	0	0	1470	6	**OK**
EmployeeCount	Number	0	1470	0	1	**ZERO VARIANCE — constant**
EmployeeNumber	Number	0	1470	0	1470	**OK**
EnvironmentSatisfaction	Number	0	1470	0	4	**OK**
Gender	Text	0	0	1470	2	**OK**
HourlyRate	Number	0	1470	0	71	**OK**
JobInvolvement	Number	0	1470	0	4	**OK**
JobLevel	Number	0	1470	0	5	**OK**
JobRole	Text	0	0	1470	9	**OK**
JobSatisfaction	Number	0	1470	0	4	**OK**
MaritalStatus	Text	0	0	1470	3	**OK**
MonthlyIncome	Number	0	1470	0	1349	**OK**
MonthlyRate	Number	0	1470	0	1427	**OK**
NumCompaniesWorked	Number	0	1470	0	10	**OK**
Over18	Text	0	0	1470	1	**ZERO VARIANCE — constant**
OverTime	Text	0	0	1470	2	**OK**
PercentSalaryHike	Number	0	1470	0	15	**OK**
PerformanceRating	Number	0	1470	0	2	**OK**
RelationshipSatisfaction	Number	0	1470	0	4	**OK**
StandardHours	Number	0	1470	0	1	**ZERO VARIANCE — constant**
StockOptionLevel	Number	0	1470	0	4	**OK**
TotalWorkingYears	Number	0	1470	0	40	**OK**
TrainingTimesLastYear	Number	0	1470	0	7	**OK**
WorkLifeBalance	Number	0	1470	0	4	**OK**
YearsAtCompany	Number	0	1470	0	37	**OK**
YearsInCurrentRole	Number	0	1470	0	19	**OK**
YearsSinceLastPromotion	Number	0	1470	0	16	**OK**

Column	Expected type	Blanks	Numeric cells	Text cells	Distinct values	Verdict
YearsWithCurrManager	Number	0	1470	0	18	**OK**
AttritionFlag	Number	0	1470	0	2	**OK**
OvertimeFlag	Number	0	1470	0	2	**OK**
Age Band	Text	0	0	1470	5	**OK**
Income Band	Text	0	0	1470	4	**OK**
Tenure Band	Text	0	0	1470	4	**OK**

3. Categorical standardisation checks

Column	Distinct values	Values found	Untrimmed cells	Case/space near-dup.	Verdict
Attrition	2	No, Yes	0	0	**OK**
BusinessTravel	3	Non-Travel, Travel_Frequently, Travel_Rarely	0	0	**OK**
Department	3	Human Resources, Research & Development, Sales	0	0	**OK**
EducationField	6	Human Resources, Life Sciences, Marketing, Medical, Other, Technical Degree	0	0	**OK**
Gender	2	Female, Male	0	0	**OK**
JobRole	9	Healthcare Representative, Human Resources, Laboratory Technician, Manager, Manufacturing Director, Research Director, Research Scientist, Sales Executive, Sales Representative	0	0	**OK**
MaritalStatus	3	Divorced, Married, Single	0	0	**OK**
OverTime	2	No, Yes	0	0	**OK**
Age Band	5	1. 18-25, 2. 26-35, 3. 36-45, 4. 46-55, 5. 56+	0	0	**OK**
Income Band	4	1. Under 3k, 2. 3k-5k, 3. 5k-10k, 4. 10k+	0	0	**OK**
Tenure Band	4	1. 0-1 yrs, 2. 2-5 yrs, 3. 6-10 yrs, 4. 11+ yrs	0	0	**OK**

4. Numeric range & logical consistency checks

Check	Min	Max	Rows failing	Verdict
Age within 18–65	18	60	0	**OK**
MonthlyIncome > 0	1,009	19,999	0	**OK**
HourlyRate > 0	30	100	0	**OK**
DailyRate > 0	102	1,499	0	**OK**
MonthlyRate > 0	2,094	26,999	0	**OK**
DistanceFromHome >= 0	1	29	0	**OK**
PercentSalaryHike > 0	11	25	0	**OK**
NumCompaniesWorked >= 0	0	9	0	**OK**
TrainingTimesLastYear >= 0	0	6	0	**OK**
YearsAtCompany >= 0	0	40	0	**OK**
YearsInCurrentRole <= YearsAtCompany	0	18	0	**OK**
YearsSinceLastPromotion <= YearsAtCompany	0	15	0	**OK**
YearsWithCurrManager <= YearsAtCompany	0	17	0	**OK**
YearsAtCompany <= TotalWorkingYears	0	40	0	**OK**
TotalWorkingYears <= Age - 18	0	40	0	**OK**
EnvironmentSatisfaction rating 1–4	1	4	0	**OK**
JobSatisfaction rating 1–4	1	4	0	**OK**
JobInvolvement rating 1–4	1	4	0	**OK**
WorkLifeBalance rating 1–4	1	4	0	**OK**
RelationshipSatisfaction rating 1–4	1	4	0	**OK**
JobLevel 1–5	1	5	0	**OK**
StockOptionLevel 0–3	0	3	0	**OK**
PerformanceRating 3–4	3	4	0	**OK**

5. New helper columns added to HR_Attrition_Data

Column	Col ref	Type	Logic	Purpose
AttritionFlag	AJ	Number (0/1)	IF(EXACT(TRIM(Attrition),"Yes"),1,0)	Average = attrition rate in any pivot; Sum = leaver count
OvertimeFlag	AK	Number (0/1)	IF(EXACT(TRIM(OverTime),"Yes"),1,0)	Average = overtime rate KPI
Age Band	AL	Text	18-25 / 26-35 / 36-45 / 46-55 / 56+	Slicer-friendly age grouping
Income Band	AM	Text	Under 3k / 3k-5k / 5k-10k / 10k+ (MonthlyIncome)	Row field for the income-band attrition chart
Tenure Band	AN	Text	0-1 / 2-5 / 6-10 / 11+ yrs (YearsAtCompany)	Row field for the tenure-band attrition chart

6. Notes & known limitations

- EmployeeCount (always 1), Over18 (always 'Y') and StandardHours (always 80) carry zero variance — excluded from all analysis. Left in the table for completeness.
- HourlyRate, DailyRate and MonthlyRate do not reconcile with MonthlyIncome in this dataset; MonthlyIncome is the only pay field used.
- All 23 range and logical-consistency checks pass — no tenure contradictions, no out-of-scale ratings, no non-positive pay values.
- Attrition is the outcome variable (16.1% of employees). It is deliberately imbalanced; rates by category should always be read alongside group size.
- Dataset is a point-in-time snapshot with no leaver dates, so no time-series or trend analysis is possible.
- Source: IBM Watson Analytics sample HR data — fictional but internally consistent, with genuine relationships between variables.

7. Build log — objects created on the Analysis sheet

Object	Location	Detail
PT_Chart_AttritionByJobRole	Analysis!A8	Rows: JobRole (sorted desc) | Avg AttritionFlag
PT_Chart_AttritionByTravelOvertime	Analysis!D8	Rows: BusinessTravel | Cols: OverTime | Avg AttritionFlag
PT_Chart_AttritionByTenureBand	Analysis!J8	Rows: Tenure Band | Avg AttritionFlag
PT_Chart_AttritionByIncomeBand	Analysis!A22	Rows: Income Band | Avg AttritionFlag
PT_Chart_LeaversByDepartment	Analysis!D20	Filter Attrition=Yes | Rows: Department | Count EmployeeNumber
PT_Chart_EngagementScores	Analysis!J22	Rows: Attrition | Avg of 5 engagement scores
CH_AttritionByJobRole	Analysis §2	Clustered bar — from pivot 1
CH_AttritionByTravelOvertime	Analysis §2	Clustered column — from pivot 2
CH_AttritionByTenureBand	Analysis §2	Line — from pivot 3
CH_AttritionByIncomeBand	Analysis §2	Column — from pivot 4
CH_LeaversByDepartment	Analysis §2	Pie — from pivot 5
CH_EngagementScores	Analysis §2	Radar — from pivot 6
PT_KPI_TotalEmployees	Analysis!A71	Count of EmployeeNumber
PT_KPI_AttritionRate	Analysis!D71	Average of AttritionFlag
PT_KPI_LeaverCount	Analysis!G71	Sum of AttritionFlag
PT_KPI_AvgMonthlyIncome	Analysis!J71	Average of MonthlyIncome
PT_KPI_OvertimeRate	Analysis!M71	Average of OvertimeFlag
PT_KPI_AvgTenureOfLeavers	Analysis!P73	Filter Attrition=Yes | Average of YearsAtCompany
SL_Department	Analysis §4	Slicer — needs Report Connections to all 12 pivots
SL_JobLevel	Analysis §4	Slicer — needs Report Connections to all 12 pivots
SL_MaritalStatus	Analysis §4	Slicer — needs Report Connections to all 12 pivots
End consumer	—	Head of Talent Retention
Perspective	—	Flight-Risk Profile — where attrition concentrates and which conditions drive it